



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

BMIT3003 DATA WAREHOUSE TECHNOLOGY

ASSIGNMENT

1. Objectives

This assignment is to assess a student's ability:

- Transform a previous relational database system into a logical star-schema architecture by correctly identifying relevant dimensions and central fact table measures.
- Implement a physically sound data warehouse design using appropriate data types, integrity constraints, and at least one Type 2 Slowly Changing Dimension table to track historical data.
- Develop the initial ETL process using SQL scripts, views, and procedures to successfully perform the first-time loading of data into the warehouse.
- Construct a subsequent loading process that incorporates data scrubbing and cleaning logic to handle dirty data and maintain data integrity.
- Present the complete assignment report in a well-organized, clear, and professional technical format following the required industry layout.

2. Learning Outcomes

Upon completion of this assignment, the students are expected to achieve:

CLO1: Apply the best practice that should be adopted in data management and data warehouse architectures that are in line with the current industry conventions. (C3, PLO6)

3. Group Formation

Students are to form a group of **THREE (3)** or **FOUR (4)** members per team from the same tutorial group by the tutor. Every member in the group is expected to contribute and participate actively in the entire process of completing the assignment. Sharing of ideas and assistance in the completion of this assignment among members is required. Students will be individually assessed and therefore, every member in the group is expected to contribute and participate actively in the entire process of completing the assignment.

IMPORTANT: *The award of marks will take into consideration the scope (i.e. complexity, correctness, clarity, and completeness) of the tasks successfully completed by the student and deliver work on schedule.*

4. Important Dates

Submission Deadlines : **Week 11 [29/8/2026] (Saturday) by 11.59PM**
 Presentation date : Week 12
 Date returned : Week 14

5. Assignment Tasks

Create a data warehouse with the use of the Star-schema modelling technique.

The data warehouse should be based on your previous/current assignment of a database system, specifically from, BMCS3183/BACS3183 Advanced Database Management or BMCS1053 Database Management course assignment.

Tasks	Marks
<u>System Overview & Background</u> Provide a brief write-up of the operational database system being transformed into the data warehouse.	-
<u>Task 1: Design of Data Warehouse</u> a) Logical Design Identify relevant dimensions and the correct measures in the central fact table b) Physical Design Write <code>CREATE TABLE</code> statements with appropriate data types, sizes, integrity constraints, and at least <i>one Type 2-dimension table</i> .	5 5
<u>Task 2: Extract, Transform, Load (ETL) Processes</u> a) Initial Loading Create <code>VIEWS</code> , <code>SELECT</code> , <code>INSERT</code> , and <code>PROCEDURES</code> to execute the first-time historical load. b) Subsequent Loading Program incremental loading scripts containing logic to handle and <i>scrub dirty data</i> .	15 15
<u>Task 3: Business Analytics Queries & Decision Support</u> Produce THREE (3) analytical reports evaluating data across multiple dimensions. Each report must include formatted charts/visualizations and deep management commentary answering 5W and 1H questions such as but not limited to: • <i>What is the data telling us?</i> • <i>Why is the change significant and what caused it?</i> • <i>What strategic actions/decisions should management take moving forward?</i>	45
<u>Task 4: Professional Report Presentation</u> Compile all tasks into a comprehensive technical report utilising standard headers, footers, numbers, and the precise sample structure.	15

6. Assignment Mark Allocations

Task	Marks	Mark Allocations
Task 1 - 2	40	Group
Task 3 - 4	60	Individual

7. The Final Assignment Report Format

The **Assessment Rubric** (refer *01-A5 AssessmentForm.xlsx*) should be submitted and uploaded together with the final assignment report during the submission. The final assignment report for all parts should contain the following items:

- Cover sheet (refer *01-A1 FrontCover.docx*)
- Plagiarism statement with student signatures (refer *01-A2 PlagiarismStatement.docx*)
- Plagiarism statement for AI generative technology (*Each student must include their own copy in the assignment report*) (refer *01-A3 PlagiarismStatement (AI Generative).docx*)
- Free Rider Policy for Student Declaration (*Each student must sign on the form to agree the tasks allocated in this assignment completion*) (refer *01-A4 Free-Rider Policy - Group Contract Form.docx*)
- Table of Contents (*organised using Table of Contents with page number in MS Word/GDocs*)
- Body of answers – all 4 tasks
- Reference section (Students are required to use APA Referencing System format)
- Appendices (if any)

You are recommended to format your report according to the following specification:

Media	Softcopy to be submitted to the <i>tutor's Google Classroom</i> .
Font Size	SQL code: 10pt (Courier New) Written description: 11pt (Times New Roman)
Font Style	Use <i>Times New Roman</i> for body text. Main headings and sub-headings should be clearly stated using suitable font styles (e.g. Arial).
Line Spacing	Typed material should be <i>single line spaced</i> .
File and folder Naming Convention	Meaningful table name, ERD, etc. Digital Submission (SQL codes, report, ERD, etc) to <i>tutor's Google Classroom</i> : Folder naming format: TutorialGroup-SystemName-StudentFullNamesWithAlphabeticalOrder (e.g., RSW1(S2)-RestaurantManagementSystem-CheahLiMei HengTzeSeong-NgSiewYongAlice)
Headers and Footers	Appropriate footers and headers should be used to enhance clarity and presentation.

Page Numbering	Ensure that all pages (except cover page) are numbered.
Report Body	• Answer each task in sequence (you must label the correct task numbers according to the questions.) • Answers for task 3 must be labelled and arranged according to individual students in ascending order of student names. • Start each new task on a new page.

8. Late Submission

All assignments should be submitted by the stated due date as **Section 4**.

Late submission of the assignment will be handled according to the Guideline for Late Submission of Coursework available at **TAR UMT's Intranet > Info Gateway > Information Gateway > Department of Examination & Credit Accumulation > Procedure > Procedure Relating to Coursework Submission**.

PART B : LATE SUBMISSION OF COURSEWORK INFORMATION

Please check/tick one of the information below :

☐	Late submission of 1 - 3 days after deadline of submission: minus 10 marks
☐	Late submission of 4 - 7 days after deadline of submission: minus 20 marks
☐	Late submission of > 7 days after deadline of submission: 0 mark

In certain circumstances, a student may be allowed to submit the assignment late with valid reason. S/he must inform the respective tutor **at least one week before** the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late, but **no guarantee** that the students will not be penalized.

9. Academic Integrity and Plagiarism

There must be originality in your work, i.e. do not copy or refer to other students. You may only work with your team members where the functionalities that you are handling depend on your team members' parts to produce the solution of this assignment. You must not share with nor refer to any part of the assignment of anyone else except your team member(s) and your tutor.

Before submitting your assignment, please make sure that you have complied with the **TAR UMT Plagiarism Policy** and **General Guidelines for Use of Generative Artificial Intelligent (AI) tools in Teaching, Learning and Assessment (TLA)**. Any cheating, attempt to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment will cause the students concerned to be penalized.

IMPORTANT: Students found to be dishonest are liable to disciplinary action.

10. Free Rider Policy

All students participating in group assessments, projects, or research-based activities at the University are required to read and adhere to the **TAR UMT Free-Rider Guidelines** (refer *03-A1 TAR UMT Policy - Free-Rider Policy.pdf*)

Students wishing to report a non-contributing member ("free-rider") within their group must formally submit the **Student Free-Rider Report Form** (refer *03-A2 Student Free-Rider Report Form.docx*).

11. Assessment Weightage

This assignment contributes **60%** of the overall coursework mark. The breakdown of the marks is in Appendix 5.

Appendix 1: Cover Page (refer *01-A1 FrontCover.docx*)

Appendix 2: Plagiarism Statement (refer *01-A2 PlagiarismStatement.docx*)

Appendix 3: Plagiarism statement for AI generative technology (refer *01-A3 PlagiarismStatement (AI Generative).docx*)

Appendix 4: Free-Rider Policy – Group Contract Form (refer *01-A4 Free-Rider Policy - Group Contract Form.docx*)

Appendix 5: Assignment Assessment Rubrics (refer *01-A5 AssessmentForm.xlsx*)

Appendix 6: APA Referencing Format (refer *02-A1 APA Referencing Format.pdf*)

Appendix 7: TAR UMT Policy – Free-Rider Policy (refer *03-A1 TAR UMT Policy - Free-Rider Policy.pdf*)

Appendix 8: Student Free-Rider Report Form (refer *03-A2 Student Free-Rider Report Form.docx*)